

Supplementary Appendix for Consensus Is Not Completion: False Convergence in Multi-Agent Discovery Workflows

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June 11, 2026

1 Purpose of the Appendix

This appendix records experimental and implementation details that are useful for reproducibility but are not required to follow the main argument. The main paper is self-contained: it defines the problem, reports the principal results, and states the limitations. The appendix expands the protocol, logging schema, split rule, and audit-policy diagnostics.

2 Closed-Loop Protocol

Each experimental state follows the same loop:

1. Fix a bounded source collection and oracle item set.
2. Run discovery agents without shared intermediate answers, or run deterministic diagnostic scanners.
3. Aggregate agent reports using consensus, standard summarization, raw union, or union-preserving summarization.
4. Compute pre-audit observable signals from reports and metadata only.
5. Apply the completion certificate without oracle labels or holdout gain.
6. If required, run an audit policy over singleton, boundary, or source-partitioned evidence.
7. Score final recall, precision, false-certification rate, audit recovery, and cost proxies against the oracle.

3 Evidence-Preserving Audit Policy Listing

The main paper describes the audit controller; the procedural listing is:

1. Run k discovery agents without shared intermediate answers, producing $Y_1, \dots, Y_k$.
2. Compute item counts $n(y) = |\{i : y \in Y_i\}|$.
3. Set conservative final set $F = \{y : n(y) \geq 2\}$ and audit queue $Q = \{y : n(y) = 1\}$.
4. Compute pre-audit confidence, overlap, source coverage, and unseen-mass signals.
5. Apply the pre-audit completion certificate.
6. If the certificate is SAFE-TO-STOP, return F with the certificate report.
7. If Q is non-empty, trigger singleton audit.
8. If confidence and overlap are high and the task has boundary-resolution risk, trigger boundary-focused holdout audit.
9. Add audit-verified singletons and holdout-discovered boundary items to F .
10. Recompute the post-audit certificate using the updated ledger and return F plus pre/post certificate reports.

4 Task Families and Oracles

Task	Source type	Oracle size	Main role
T1	synthetic code repository	35	aggregation-loss controls
T2	synthetic policy documents	30	blind-spot controls
T3	Click repository snapshot	149	real-repo deprecation audit
T4	Requests repository snapshot	304	real-repo TLS audit
T5	itsdangerous repository snapshot	160	external timed-signing audit

Table 1: Task families used by the current paper.

5 Oracle Construction Protocol

Each task uses a bounded source snapshot and a released oracle file under `experiments/false_convergence_pilot/results/`. Oracle labels are used only for offline scoring, not during blind agent generation or pre-audit certification.

Task	Oracle file	Commit / source	Size
T1	T1_hard_repo_oracle.json	synthetic AcmePay repo	35
T2	T2_policy_docs_oracle.json	synthetic policy docs	30
T3	T4_real_repo_click_deprecation_oracle.json	Click 8a1b1a3	149
T4	T5_real_repo_requests_tls_oracle.json	Requests 1190afd	304
T5	T6_real_repo_itsdangerous_timed_signing_oracle.json	itsdangerous 672971d	160

Table 2: Oracle artifacts used for scoring.

Inclusion criteria. T1 includes code, configuration, registry, fallback, and replay locations that resolve to AcmePay v1. T2 includes policy cases whose state, flow, and service resolution imply AcmePay v1 handling. T3 includes Click line-level items in the deprecated API surface audit, including implementation, warning emission, help formatting, validation guards, tests, bounded changelog entries, and direct user documentation. T4 includes Requests line-level items for TLS verification, CA-bundle resolution, client certificates, TLS/SSL error handling, tests, test infrastructure, and direct user documentation. T5 includes itsdangerous timestamped signing, max-age expiration, exception, serializer, test, documentation, export, and changelog lines.

Exclusion criteria. We exclude broad file-level answers, incidental keyword mentions outside the task boundary, unrelated uses of terms such as `verify`, and files outside the bounded task snapshot. For T3–T5, oracle files include an `oracle.policy` field describing these boundaries.

Construction process. The current oracles were built from task definitions, keyword/static scans, source inspection, and manual normalization to line-level items with evidence spans and buckets where available. The current paper treats them as closed-world evaluation artifacts, but we do not claim independent double annotation. Before submission, the real-repository oracles should receive an independent second pass that records reviewer, changed items, ambiguous cases, and final resolution rules.

6 v2 Diagnostic Logging Schema

The v2 offline diagnostic writes a candidate/action log with the following fields:

```
run_id, repository, task_family, seed, agent_id, model,
prompt_variant, round_id, query_or_action, source_file,
source_region, candidate_item, first_seen_round, support_count,
confidence, token_input, token_output, tool_calls, wall_clock
```

The run generated 2,970 runs, 1,572,201 candidate/action rows, 2,340 pre-audit states, 439 safe states, and 1,901 unsafe states. Oracle labels are used only after candidate generation for residual missing mass, calibration, and evaluation.

7 Split Rule

Train, calibration, and test splits are grouped by (repository, task family, seed). No derived state from a group crosses splits. The current split uses seeds 1–2 for training, seed 3 for calibration, and seeds 4–5 for test. This prevents near-duplicate states from the same seed from appearing in both calibration and test.

8 Audit Policy Diagnostics

Policy	States	Pre recall	Post recall	Recovered TP
no audit	936	.267	.267	0
boundary holdout	936	.267	.283	2,596
random holdout	936	.267	.306	7,026
risk-triggered audit	936	.267	.535	51,548
source-partitioned audit	936	.267	.535	51,552
always holdout	936	.267	1.000	121,869

Table 3: v2 held-out audit-policy summary. The deterministic offline holdout is a diagnostic stress test and should not be read as an online LLM audit result.

8.1 Source-Aware Audit Prototype

The T3 certificate tells the workflow not to stop, but a practical system also needs an audit policy that converts retained evidence into recovery. We evaluate an offline source-aware audit prototype over G3/holdout candidates, using deterministic task-policy predicates over source lines. This is an audit-policy upper bound, not a blind LLM result. On T3, the raw candidate pools average 0.998 recall but only 0.660 precision. Source-aware filtering keeps the same average recall and raises precision to 1.000; a bounded source sweep reaches 1.000 recall and 1.000 precision. On T4, candidate filtering reaches 0.849 recall and 1.000 precision, showing that precise audit is possible but search coverage remains insufficient.

8.2 Per-Seed Audit Behavior and Cost Proxy

The audit policy exposes rather than hides uncertainty. In T1 seed01, it audits and verifies 13 singleton true positives. In T2 seed01, it verifies 2 singleton true positives. In T2 seed02/03, there are no singletons, but high agreement and high confidence trigger boundary-focused holdout, recovering 2 new items in each seed. In T1 seed02/03, the audit policy leaves 4 and 5 singleton items in the audit queue and does not mark completion, avoiding raw union’s false positives. In T3 seed01, the evidence-preserving audit policy reaches 0.993 recall and 0.955 precision by auditing singleton evidence. In seed02 and seed03, however, the available holdout scout is itself incomplete, so the audit policy recovers only to 0.805 and 0.832 recall. The completion certificate catches

this boundary: for all three T3 seeds it returns UNSAFE-TO-STOP, triggered by low mean confidence, singleton missing-mass signals, and Chao unseen-mass estimates.

Removing singleton audit leaves the aggregation-loss singletons in T1 seed01 and T2 seed01 unrecovered. Removing the common-blind-spot trigger leaves T2 seed02/03 at 28/30 recall despite perfect G3 agreement. Removing holdout converts hidden risk into a REQUIRES-AUDIT state but cannot recover missing items. Thus the two triggers correspond to distinct failure mechanisms, and holdout is the mechanism that turns retained risk into verified recovery.

The offline audit-policy stress-test logs do not contain reliable token or wall-clock accounting, so we report a proxy cost: audit queue size plus one unit per triggered holdout. Under this proxy, T1 seed01 uses 14 audit actions to recover 13 true positives, T2 seed01 uses 3 actions to recover 2 true positives, and T2 seed02/03 use one holdout unit to recover 2 true positives each. In T1 seed02/03, the audit policy avoids the 4 and 5 false positives introduced by raw union, but correctly refuses to certify completion without further audit.

9 v2 Leakage and Generalization Audit

The v2 audit reads the frozen offline diagnostic table and writes new outputs to `experiments/false_convergence_pilot/v2_audit_outputs/`. The declared v2 feature set excludes oracle and label columns (`recall`, `true_positive`, `residual_missing_mass`, and `unsafe`), task/repository identifiers, split/seed identifiers, run ids, post-audit fields, and cost columns. All audited rows are pre-audit states. A metadata-only shortcut probe is reported separately and is not used as a valid method.

Setting	AUROC	FCR	Safe coverage
v2 in-distribution test	.999	.017	.966
metadata-only probe	.988	–	.000
leave-one Click	.591	–	.000
leave-one policy docs	.546	–	.000
leave-one Requests	.695	–	.000

Table 4: Frozen v2 audit. Held-out repository/task-family validation fails to certify any state under the calibration criterion.

10 Online Requests Audit-Controller Evaluation

We ran the P0 Requests TLS online audit-controller loop using `gpt-5.3-codex`, seeds 04–08, search budget 180, max 1200 lines per file, and the fixed Requests commit from the main paper. Outputs are stored in `experiments/false_convergence_pilot/online_audit_controller/T5_requests_tls/`. The online suite contains 60 blind discovery calls and 45 audit calls: five boundary-focused holdout agents plus 40 random/singleton candidate-verifier agents. The audit-policy result

manifest is stored at `experiments/false_convergence_pilot/online_audit_controller/T5_requests_tls/audit_policy_eval/MANIFEST.json`.

Condition	N	Jaccard	Consensus R	Union R	Precision	Tokens
homogeneous	5	.813	.678	.732	.710	213k
prompt-diverse	5	.778	.662	.724	.709	213k
source-partitioned	5	.000	.000	.724	.737	74k
independent-context	5	.285	.626	.732	.729	143k

Table 5: Online discovery grid before post-audit evaluation. Source partitioning keeps union recall comparable but makes vote-only consensus unusable.

Policy	Pre R	Post R	Precision	Rec. TP	Intro. FP	Tokens
no audit	.491	.491	.829	0.0	0.0	0k
random holdout	.491	.554	.755	18.9	11.9	17k
singleton audit	.491	.719	.725	69.1	38.7	12k
boundary holdout	.491	.694	.755	61.7	24.6	71k
source-partitioned review	.491	.733	.730	73.3	38.5	55k
risk-triggered audit	.491	.733	.713	73.3	45.5	83k
always holdout	.491	.753	.706	79.5	51.1	163k

Table 6: Online audit-policy means over 20 seed/configuration states. Tokens are additional audit tokens.

The full per-seed/per-condition logs are stored in `audit_policy_eval/ONLINE_AUDIT_POLICY_RESULTS.csv` and `audit_policy_eval/ONLINE_AUDIT_POLICY_RESULTS.md`. Discovery-grid means are stored in `ONLINE_DISCOVERY_GRID_SUMMARY.csv` and `ONLINE_DISCOVERY_GRID_SUMMARY.md`. These results close the Requests P0 method-effect loop, but they also show a remaining gap: even always-holdout averages only .753 recall, so the current controller should continue searching or auditing rather than certify completion.

11 Held-Out Click Staged Controller

After the Requests development experiment, we froze a staged controller: conservative consensus, singleton verification, source-partitioned review when singleton mass/source gaps remain, and abstention whenever escalation was needed. We then evaluated it on held-out online Click runs with seeds 04–06 and three discovery configurations. Outputs are stored in `experiments/false_convergence_pilot/online_heldout_click_staged_controller/`.

These held-out results support a narrower claim than automatic completion: the staged controller improves recall–cost tradeoff while preserving abstention when source escalation is required. It does not yet solve safe-to-stop coverage.

Policy	Post R	Prec.	Rec. TP	Intro. FP	Audit tok.	E2E tok.	FCR	Abstain
no audit	.465	.769	0.0	0.0	0k	166k	.000	1.000
singleton audit	.722	.630	38.2	29.6	9k	175k	.000	1.000
source-partitioned review	.755	.577	43.2	44.8	57k	223k	.000	1.000
staged controller	.761	.555	44.0	54.1	56k	223k	.000	1.000
always holdout	.768	.492	45.1	80.3	177k	343k	.000	1.000

Table 7: Held-out Click online results over nine seed/configuration states. The staged controller nearly matches always-holdout recall at about one third of the audit-token cost, but abstains on every state. Safe coverage is therefore 0.000, while false certification is also 0.000.

12 External itsdangerous Staged Controller

The external validation repository is `pallets/itsdangerous` at commit `672971d66a2ef9f85151e53283113`. The staged controller, prompt, budget, and thresholds were frozen before this task was scored. Outputs are stored in `experiments/false_convergence_pilot/online_external_itsdangerous_staged_controller/`.

Policy	Post R	Prec.	F1	Rec. TP	Intro. FP	Audit tok.	E2E tok.	Abstain
no audit	.569	.841	.679	0.0	0.0	0k	31k	1.000
singleton audit	.867	.719	.786	47.8	26.1	9k	40k	1.000
source-partitioned review	.871	.733	.796	48.3	22.1	11k	43k	1.000
staged controller	.879	.709	.785	49.7	29.6	19k	50k	.889
always holdout	.890	.675	.768	51.4	39.9	41k	72k	1.000

Table 8: External itsdangerous online results over nine seed/configuration states. The staged controller has one certified unsafe state, so its false-certification rate among certified states is 1.000 and safe coverage is 0.000.

The safe-state sweep on seed04 added three boundary-focused agents with a larger search budget. The base union had recall .894 and precision .641. The boundary-augmented union had recall .900 and precision .634, recovering one true positive and introducing three false positives at 39,602 additional audit tokens. The sweep therefore did not produce a real online safe state.

13 Bootstrap and Paired Tests

The script `experiments/false_convergence_pilot/tools/analyze_staged_controller_stats.py` computes seed-clustered bootstrap intervals over Requests, Click, and itsdangerous. Requests is included in policy means but excluded from staged paired tests because the Requests development run did not include a frozen staged-controller arm.

Comparator	Recall diff	Precision diff	F1 diff	Audit-token diff
singleton audit	.025 [.012,.039]	-.042 [-.067,-.019]	-.013 [-.025,-.002]	28k [12k,47k]
source review	.007 [.005,.009]	-.023 [-.029,-.017]	-.012 [-.017,-.009]	3k [-6k,9k]
always holdout	-.009 [-.013,-.005]	.048 [.034,.063]	.029 [.018,.039]	-71k [-111k,-37k]

Table 9: Paired staged-controller differences over Click and itsdangerous. The intervals are descriptive seed-clustered bootstrap 95% intervals and should not be used to retune the frozen controller.

14 Oracle Review and Replication Status

The real-repository oracle review status is logged at `experiments/false_convergence_pilot/oracle_reviews/ORACLE_SECOND_REVIEW_STATUS.md`. No independent human second-review diff has been produced yet, so the paper does not claim double annotation. Second-model replication is also not yet run; it remains a TODO until a second model, fixed prompt, manifest, logs, and scores are present.

15 SeekerGym Status

The SeekerGym local checkout/schema is not present, so no benchmark result is claimed. The manifest is stored at `experiments/false_convergence_pilot/benchmark_runs/seekergym/MANIFEST.json` and contains only TODOs for schema mapping, oracle construction, trace conversion, and cost logging.

TRQA and WideSearch are likewise not run in this repository. They should remain TODO entries until a fixed subset, oracle mapping, run manifest, logs, and scoring outputs exist.

16 Additional Figures

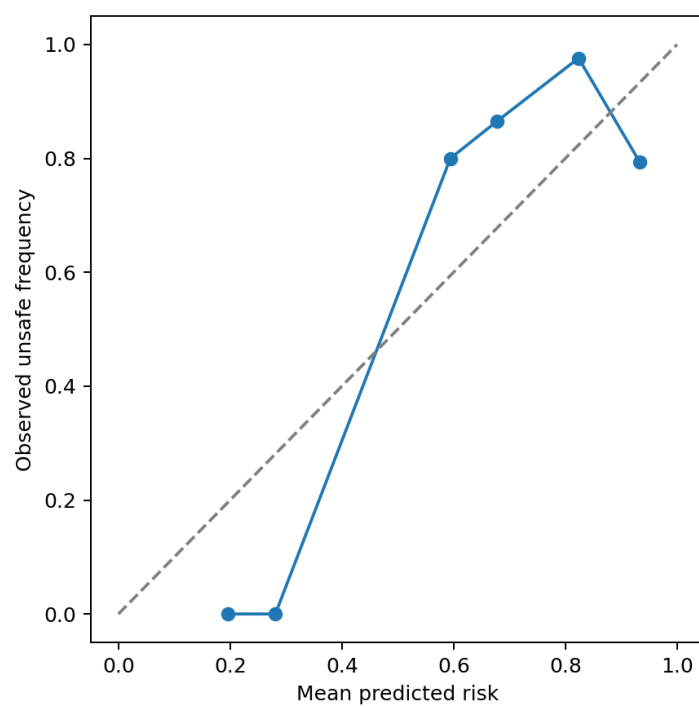


Figure 1: Calibration curve for the v1 risk proxy in the v2 diagnostic suite.